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OpenShift Installation

Platforms, Prerequisites, and Install Methods

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Agenda

- Supported platforms
- Prerequisites and requirements
- Networking and DNS
- IPI vs UPI vs Assisted Installer
- Step-by-step: AWS (cloud)
- Step-by-step: Bare Metal
- Air-gapped installation options

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Supported Platforms

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Installation Platforms

Cloud, on-premise, hybrid

- AWS, Azure, GCP (public cloud)
- VMware vSphere
- Red Hat OpenStack Platform
- Nutanix
- Bare metal (IPI, UPI, Assisted)
- IBM Power, IBM Z
- Oracle Cloud Infrastructure
- Agent-based for edge/air-gapped

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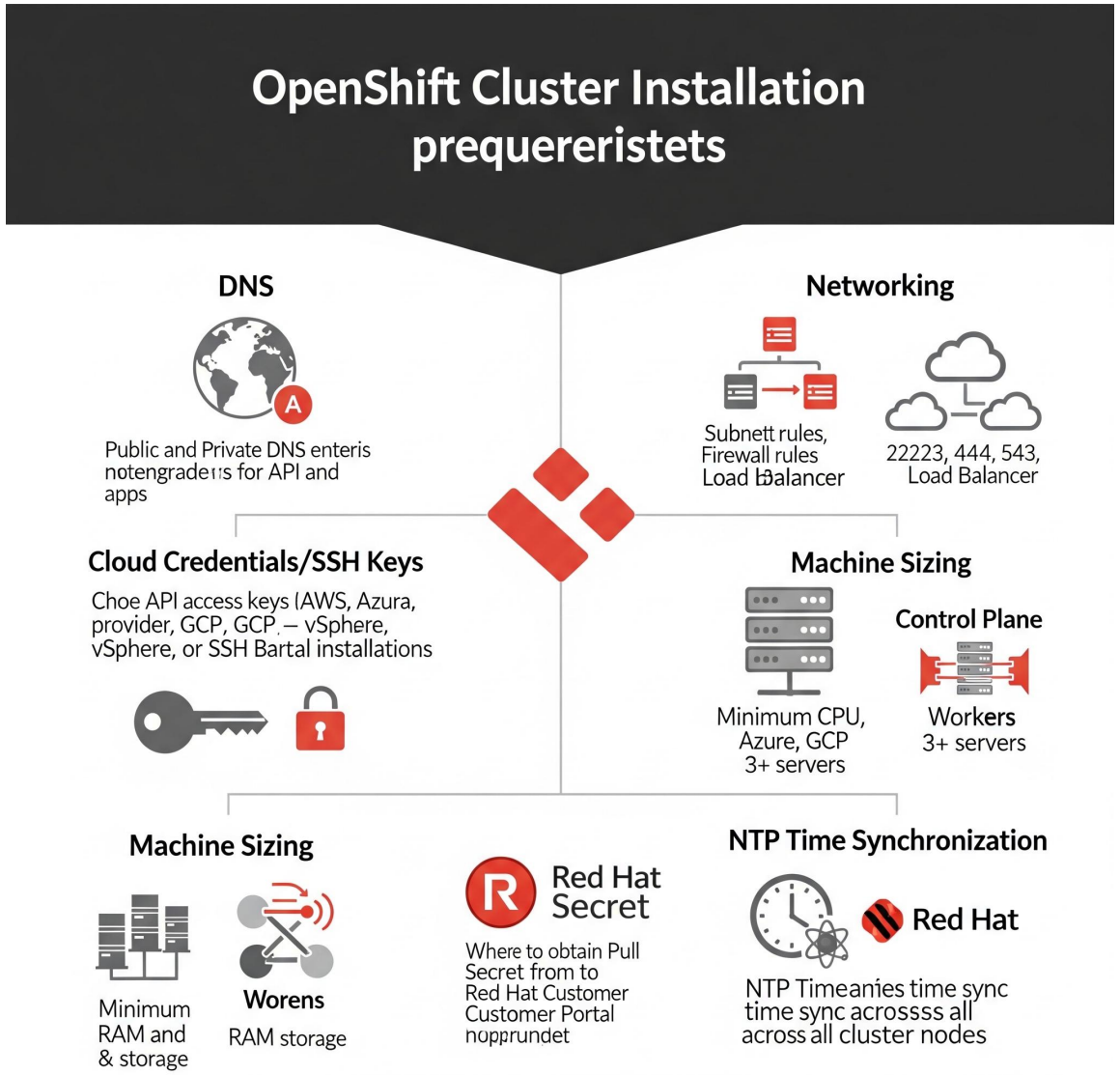
Prerequisites



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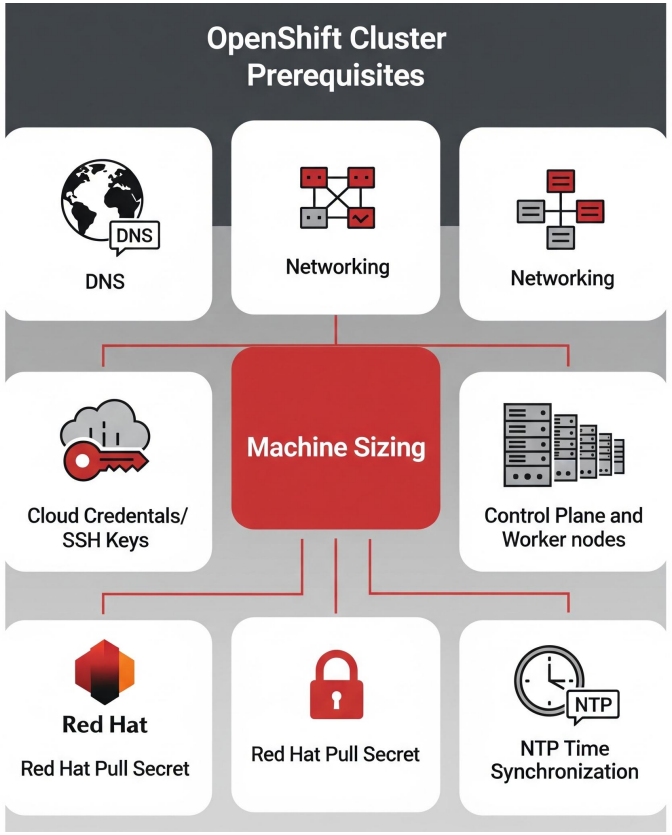
Prerequisites Overview

Common requirements across platforms

- Base domain and DNS access
- Network: routable IPs, no conflicts
- Credentials: cloud API keys or SSH keys
- Machine sizing: Control plane + workers
- Pull secret from Red Hat (cloud.redhat.com)
- Time synchronization (NTP) on all nodes



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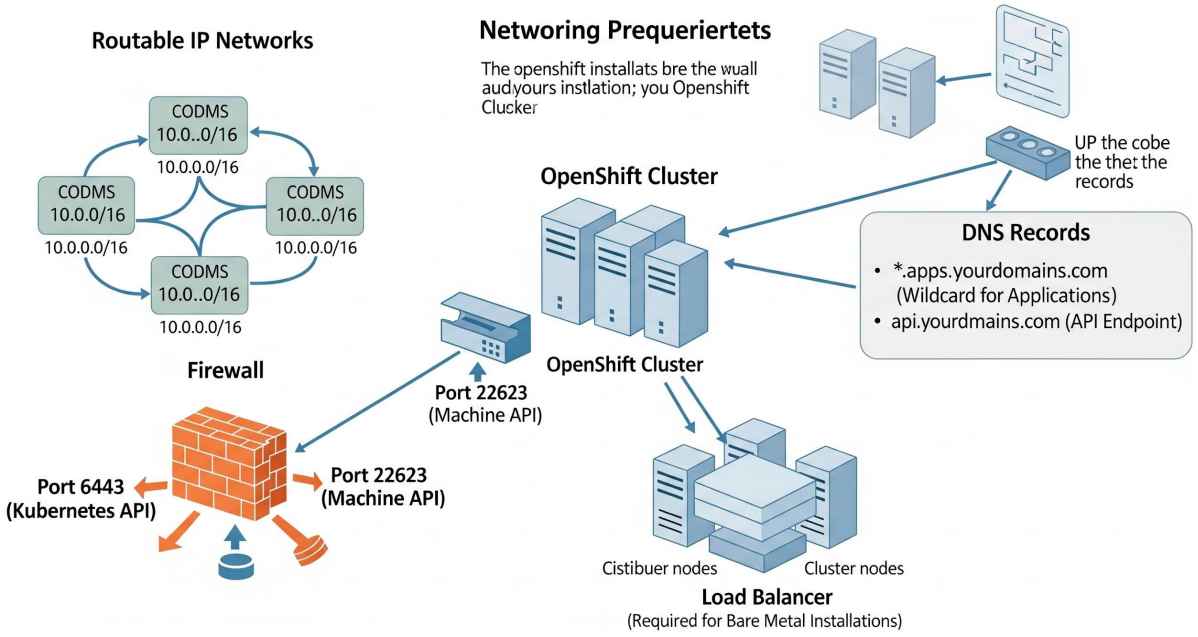
Networking & DNS

Networking Requirements

Critical for cluster communication

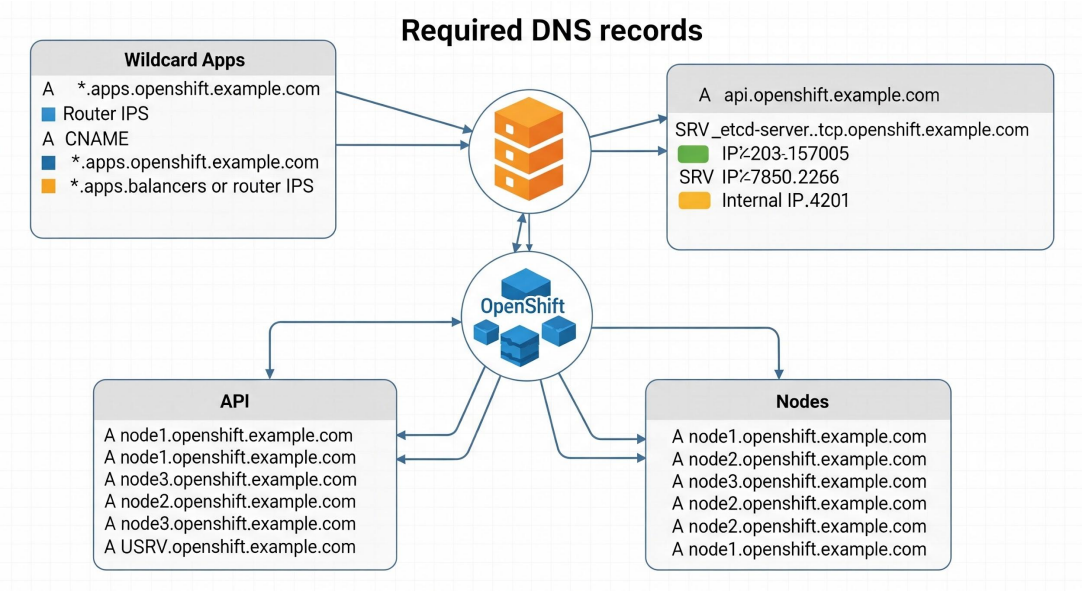
- Routable network for API, ingress, nodes
- No overlap with existing networks
- Load balancer for API and ingress (bare metal)
- Firewall: allow API 6443, machine config 22623
- MTU and bandwidth: check cluster size



DNS Configuration

Base domain and records

- Wildcard for apps: *.apps.<cluster>.<domain>
- API: api.<cluster>.<domain> → control plane LB
- API internally: api-int.<cluster>.<domain>
- Bootstrap, master, worker node records
- Reverse DNS (PTR) for bare metal



IPI vs UPI vs Assisted Installer

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Installation Methods Compared

IPI · UPI · Assisted

IPI – Installer Provisioned
Full automation
Cloud focus (AWS, Azure, GCP)
Installer creates VMs, LB, DNS
Fastest, least flexible

UPI – User Provisioned
You create infrastructure
Any platform, any layout
Manual DNS, LB, nodes
Max flexibility

Assisted Installer
Web UI, no bootstrap node
Bare metal, vSphere, Nutanix
Pre-flight validation
Easier than UPI, flexible

Source: Red Hat OpenShift documentation

Step-by-Step: AWS (Cloud)

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AWS Installation (IPI)

Installer Provisioned Infrastructure

- 1. Create AWS account, configure IAM
- 2. Get pull secret from cloud.redhat.com
- 3. Download openshift-install for your platform
- 4. Create install-config.yaml (base domain, region, credentials)
- 5. Run: openshift-install create cluster
- 6. Installer provisions VPC, subnets, machines
- 7. Kubeconfig saved to ./auth/kubeconfig

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AWS Post-Install

Verify and access

- Console URL printed at end of install
- Default admin / kubeadmin password in auth/kubeadmin-password
- Configure OC login and start deploying
- Add identity provider (LDAP, OIDC) for production

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Step-by-Step: Bare Metal

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Bare Metal Installation

Options: UPI, Assisted, or Agent-based

- 1. Provision servers: 3 control plane + workers
- 2. Configure DHCP (or static) and PXE boot
- 3. Set up load balancer for API/ingress
- 4. Configure DNS (api, *.apps, bootstrap, nodes)
- 5. Create manifest + ignition configs (UPI) or use Assisted UI
- 6. Boot bootstrap node, then control plane, then workers
- 7. Remove bootstrap after cluster is up

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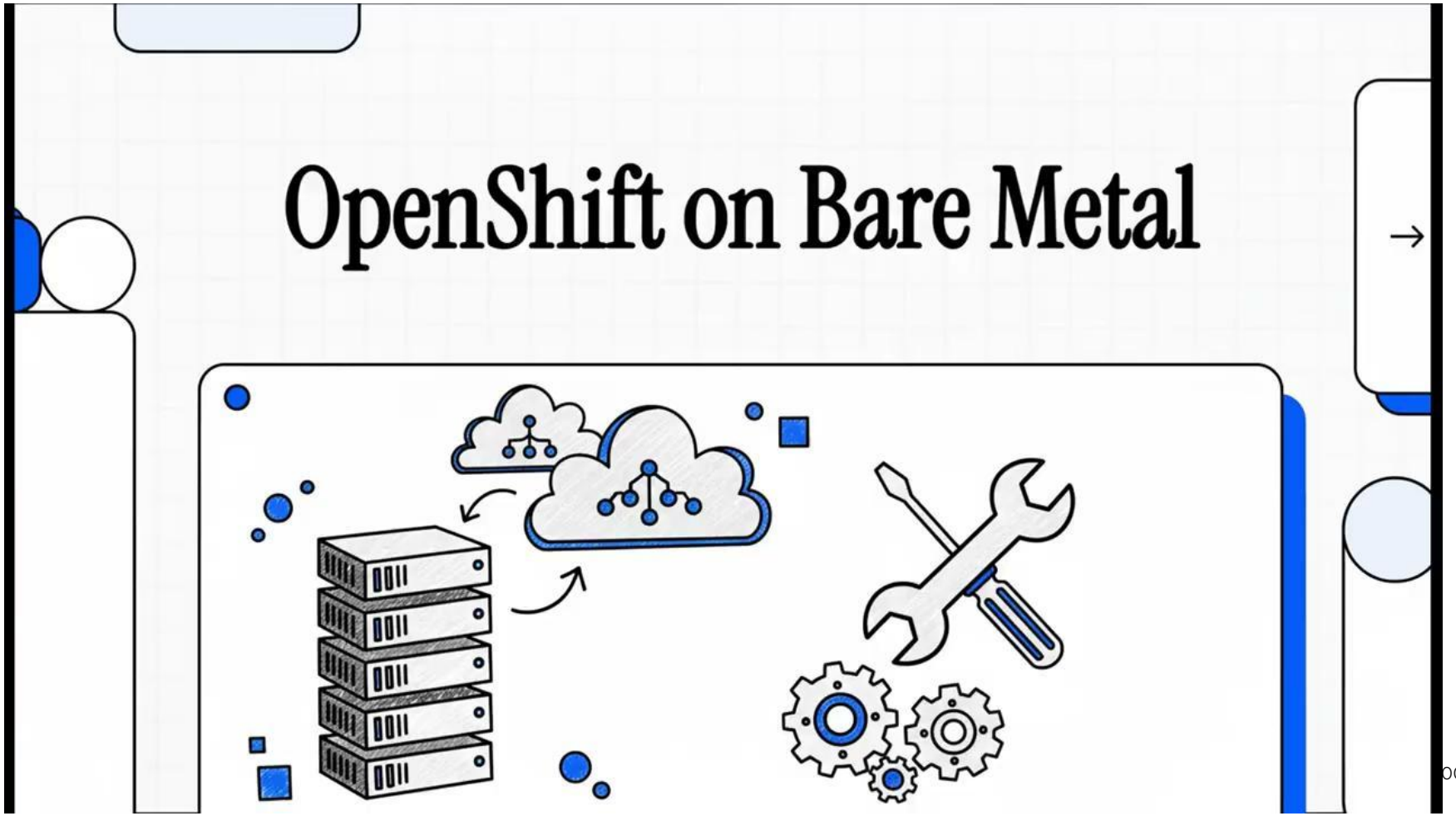
Bare Metal: Assisted vs UPI

Ease vs control



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Air-Gapped Installation

Air-Gapped / Disconnected Options

Install without public registry

- Mirror OpenShift images to internal registry
- Use oc adm release mirror –to <local-registry>
- Provide pulls from internal registry in install-config
- Agent-based installer supports disconnected with local mirror
- Assisted Installer: limited; Agent-based: full support

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Air-Gapped Workflow

From connected to disconnected

- 1. On connected host: mirror release and operator images
- 2. Export mirror content to portable media
- 3. Import into air-gapped environment
- 4. Configure install to use local registry
- 5. Run installer (Agent-based or UPI)

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Thank You

Questions? See docs.openshift.com for detailed guides.

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